Do System One Models Make Better Rerankers? Decision vs. Reranker Fine-Tuning on One Backbone

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Abstract. System One models – decision models that answer a question about a state with a calibrated probability – have been proposed as rerankers, and open decision fine-tunes now match the closed original. On six Qwen backbones (0.6B–27B) and ten collections, we compare the zero-shot base model, its decision fine-tunes (scoring candidates jointly or in isolation) and its reranker fine-tunes, including one we trained. Reranker fine-tuning wins on every backbone where both exist, also on collections without training data, while decision fine-tuning stays within a few points of the base model. Nor are decision models consistently better calibrated: reranker fine-tunes give better sets at a fixed probability cutoff, and ours, trained with a pointwise loss on its yes/no logit, is the best calibrated system we tested.

Keywords: Reranking · Calibration · System One Models · Fine-Tuning

1 Introduction

System One models, or decision models, answer a yes/no question about a structured state with a calibrated probability, named after the fast, intuitive mode of thinking [14]. They are marketed widely, and public experiments show that the closed Jev reranks BM25 candidates as well as strong cross-encoders when asked whether each candidate is relevant [27,13,15,11]. Open decision fine-tunes of Qwen models (Clef [4], Kev [23]) followed within weeks [FLAG release dates of Jev, Clef and Kev], and in our experiments Kev-27B matches Jev as a reranker (Sect. 4). That points not to a breakthrough behind a closed model, but to a training paradigm – decision fine-tuning with RLCD [FLAG cite and expand RLCD] – that anyone can apply to an open backbone, and whose merits for retrieval are untested.

The fair question is therefore decision fine-tuning vs. the established alternative, reranker fine-tuning, on the same backbone. We compare *triplets*: the base model asked zero-shot, its decision fine-tune (scored jointly or in isolation) and its reranker fine-tune, on six Qwen backbones from 0.6B to 27B parameters and ten collections. Since calibrated probabilities are the decision models' distinctive feature, we also ask whether they at least deliver that:

RQ1 On the same backbone, does decision fine-tuning make a better reranker than reranker fine-tuning or no fine-tuning?

RQ2 Are decision models at least better calibrated than reranker fine-tunes of the same backbone?

Both answers are negative. (1) On every backbone where both exist (0.6B–9B), reranker fine-tuning beats decision fine-tuning, also on the collections without training data; decision fine-tuning stays within a few points of the untuned base model, and scoring all candidates jointly only helps models trained on long states. (2) Decision fine-tunes are not consistently better calibrated than reranker fine-tunes of the same backbone and give no better set at a fixed cutoff; our own Qwen3.5-9B reranker fine-tune, trained with a pointwise loss on its yes/no logit, ranks like the best published one and is the best calibrated system we tested. Code and raw outputs are available.[1]

2 Related Work

Fine-tuned rerankers. Cross-encoders fine-tuned on MS MARCO [20,21] and their LLM successors read relevance from the logit of a relevance token or a scoring head; recent pointwise LLM rerankers fine-tune open backbones such as LLaMA [17], Qwen3 (Qwen3-Reranker [35]) and Qwen3.5 (Prism [34]). Listwise rerankers output a ranking instead [30,25]; the Set-Encoder lets passages attend to each other [28]; Abdallah et al. [1] compare many of them on our candidates.

System One models in retrieval. Decision models are trained to answer questions about a state with a probability (decision fine-tuning, RLCD [FLAG cite]); open fine-tunes exist as Clef [4] and Kev [23,22], and Laya is another open System One model [5]. Their use as rerankers has so far been studied only in non-peer-reviewed reports: S1Rank and jev-rerank score BM25 candidates with Jev [27,13], blog posts report on Jev as a reranker [15,11,32], and Laya has been used as a relevance gate before recall [29]. None compares decision fine-tuning with reranker fine-tuning on the same backbone.

Calibration and cutoffs. Modern neural networks tend to be overconfident, and temperature scaling is a simple post-hoc fix [12]; neural rankers are no exception [24]. Calibrated probabilities matter wherever a score is read on an absolute scale, e.g. to truncate a ranked list [2].

3 Experimental Setup

Triplets. On each of six backbones (Qwen3-0.6B/8B, Qwen3.5-0.8B/4B/9B, Qwen3.8-27B) we compare three objectives. *None*: the post-trained release, asked the decision models' question in its chat format (thinking off), scored by $P(\text{yes})$

[1] [TODO: anonymised link]

at the first answer token. *Decision fine-tuning*: Kev [23,22] (LoRA on the -Base checkpoints up to 9B, a full fine-tune of the post-trained 27B), Clef-Flash (9B) and Clef (27B) [4]. *Reranker fine-tuning*: Qwen3-Reranker [35] (on the -Base checkpoints Kev starts from) and Prism [34], scored by $P(\text{yes})$ over the "yes"/"no" logits; none is public for 27B. On Qwen3.5-9B we add our own (*Ours-9B*), trained without decision data: LoRA ($r = 32$) on the post-trained release, one epoch over 100k MS MARCO training queries with one positive and 15 hard negatives each, InfoNCE plus binary cross-entropy on the yes/no logit, training queries matching any test topic removed, checkpoint chosen on 200 MS MARCO dev queries. Qwen3-Reranker, Prism and Ours-9B saw MS MARCO; Kev's sources contain neither MS MARCO nor BEIR [22].

Collections and candidates. TREC DL 2019–2023 [9,6,7,8,10] (MS MARCO v1/v2 passages [19]) and TREC-COVID, NFCorpus, SciFact, Touché and DBPedia from BEIR [31]; TREC-COVID, Touché and DBPedia have no training split in any fine-tune's data and form the *zero-shot subset* (ZS). Every system reranks Pyserini's BM25 top-100 [16] (official top-100 files for DL 2021–2023, the same set), the candidates of S1Rank [27], RankGPT [30], RankZephyr [25] and Abdallah et al. [1], whose published numbers we use where we did not recompute them.

Decision models. All decision models get S1Rank's question [27], fixed in advance ("Is document `{doc}` relevant to the `query`? Relevant means it answers the query or contains information that directly addresses it."), with a state holding all 100 candidates (*joint*) or one (*isolated*); both modes are pointwise. Besides the triplets we run Jev 1.13.0 (closed), Laya [5] (isolated only; its context is too short), monoT5-3B [21], bge-reranker-v2-m3 [3], the Set-Encoder [28] and RankZephyr (DL only).

Metrics. nDCG@10 (ir_measures [18]), with paired t-tests over queries, Holm-corrected across the ten collections ($p < 0.05$); expected calibration error (ECE, 15 bins) [12] on the judged pairs of DL 2019–2023 (relevant from grade 2), the only collections with judged non-relevant passages. *One cutoff*: keep every candidate with $p \geq 0.5$; set F_1 and its regret against the best cutoff per collection (unjudged = non-relevant). *After the fact*: temperature and Platt scaling of each reranker's logit, fitted on the BM25 top-100 of 1,000 MS MARCO dev queries. *Sanity checks*: our BM25 matches Pyserini's reproductions [26], monoT5-3B and RankZephyr are within 0.2 points of their published DL 2019/2020 numbers [1], Jev joint within 0.5 of S1Rank's [27] on four of five shared collections (1.5 below on TREC-COVID).

4 Decision vs. Reranker Fine-Tuning

Reranker fine-tuning wins on every backbone. On all five backbones where both exist, the reranker fine-tune ranks better than the decision fine-tune in either

Table 1. Base model, decision fine-tune and reranker fine-tune on the same backbone (nDCG@10, ×100, BM25 top-100). Base: the post-trained release asked S1Rank's question zero-shot. Decision fine-tunes in joint and isolated mode. Avg: mean over all ten collections; ZS: mean over the collections without a training split (TREC-COVID, Touché, DBPedia). Best per backbone and column in bold. Paired t-test, Holm-corrected over the ten collections, $p < 0.05$: * reranker fine-tune vs. the decision fine-tune with the best Avg, † that decision fine-tune vs. the base model, ‡ joint vs. isolated. No reranker fine-tune of the 27B backbone exists.

			TREC DL					BEIR						
System	Training	Mode	DL19	DL20	DL21	DL22	DL23	COVID	NFC	SciF	Touché	DBP	Avg	ZS
Qwen3-0.6B														
Qwen3-0.6B	base		32.6	30.1	32.5	17.3	14.3	50.3	16.2	18.8	11.5	14.3	23.8	25.4
Kev-0.6B	decision	joint	18.4‡	13.9‡	19.1‡	12.3‡	11.0‡	34.0‡	12.0‡	1.1‡	4.5‡	6.1‡	13.2	14.9
Kev-0.6B	decision	isolated	33.4	29.3	39.8	22.5	18.8	58.2	17.0	16.2	13.0	21.5†	27.0	30.9
Qwen3-Reranker-0.6B	reranker		**69.5***	**64.0***	**65.2***	**46.4***	**38.8***	**86.2***	**36.4***	**76.6***	**35.7***	**42.7***	**56.1**	**54.9**
Qwen3-8B														
Qwen3-8B	base		70.4	64.9	65.8	41.4	42.1	80.2	34.1	72.8	24.0	37.1	53.3	47.1
Kev-8B	decision	joint	63.4‡	57.7	60.1	37.5	35.7	48.2‡	19.8‡	19.2‡	9.7‡	36.7‡	38.8	31.5
Kev-8B	decision	isolated	71.2	62.1	61.4†	38.6	37.6†	80.8	35.2	73.9	26.5	40.0†	52.7	49.1
Qwen3-Reranker-8B	reranker		**73.2**	**66.8**	**67.0***	**51.2***	**45.2***	**84.0**	**37.3***	**79.9***	**43.1***	**46.2***	**59.4**	**57.8**
Qwen3.5-0.8B														
Qwen3.5-0.8B	base		47.7	42.2	53.3	30.2	27.9	65.3	20.4	41.1	21.9	25.8	37.6	37.7
Kev-0.8B	decision	joint	35.7	32.2	38.6‡	19.9‡	18.2‡	45.3‡	18.6	20.2‡	9.4‡	14.6‡	25.3	23.1
Kev-0.8B	decision	isolated	42.8	39.8	48.3	25.8	23.8	71.9	20.9	36.0	18.0	32.2†	36.0	40.7
Prism-0.8B	reranker		**72.3***	**67.7***	**66.3***	**47.0***	**43.7***	**84.2***	**36.1***	**76.3***	**35.6***	**41.7***	**57.1**	**53.8**
Qwen3.5-4B														
Qwen3.5-4B	base		69.0	63.1	66.2	40.5	40.1	79.5	34.1	72.9	24.8	37.0	52.7	47.1
Kev-4B	decision	joint	65.5	62.9	62.4	37.7	38.3	75.3‡	29.3‡	62.4‡	23.7	40.7	49.8	46.6
Kev-4B	decision	isolated	62.6†	63.4	62.0	39.9	35.4	82.6	34.1	70.7	23.0	39.7†	51.3	48.4
Prism-4B	reranker		**74.4***	**68.9***	**70.3***	**50.1***	**46.2***	**84.6**	**37.4***	**76.9***	**34.3***	**43.7***	**58.7**	**54.2**
Qwen3.5-9B														
Qwen3.5-9B	base		69.6	65.1	67.7	42.9	38.1	81.2	33.6	75.0	27.5	35.8	53.7	48.2
Kev-9B	decision	joint	63.1	65.3	63.6	38.9	39.1	78.2	31.5	60.9‡	22.6	38.5‡	50.2	46.4
Kev-9B	decision	isolated	65.7	63.4	60.8	38.6	35.4	77.7	32.8	72.6	19.8	34.1	50.1	43.8
Clef-Flash	decision	joint	68.8	67.3	64.3	44.2	40.4	82.1	35.1‡	65.4	21.3	38.8‡	52.8	47.4
Clef-Flash	decision	isolated	68.8	69.9†	66.4	45.5	43.8†	82.3	**37.3**†	68.7†	22.1	42.6†	54.7	49.0
Prism-9B	reranker		**74.0***	**71.1**	**71.0***	51.6*	44.2	83.0	37.2	**79.8***	34.7*	43.2	59.0	53.7
Ours-9B	reranker		73.8*	70.0	67.3	**52.1***	**47.0**	**84.2**	37.1	77.9*	**36.5***	**47.7***	**59.4**	**56.1**
Qwen3.8-27B														
Qwen3.8-27B	base		71.6	69.0	68.5	46.2	44.1	83.0	37.6	78.5	30.2	45.0	57.4	52.7
Kev-27B	decision	joint	**73.9**	**73.3**†	69.9	51.1†	**49.3**†	**85.9**†	**38.7**†	**81.1**†	**40.8**‡†	46.9†	**61.1**	**57.9**
Kev-27B	decision	isolated	73.4	71.0	**70.1**	**51.3**	46.8	85.5	38.3	79.9	34.0	**47.5**	59.8	55.7
Clef	decision	joint	71.4	70.1	68.6	49.3	45.9	82.9	36.3	62.9‡	30.8	42.8	56.1	52.2
Clef	decision	isolated	69.4	69.4	66.7	44.1	42.9	83.5	37.8	71.9	33.9	44.1	56.4	53.8

mode (Table 1; average over ten collections, Qwen3-0.6B: 56.1 vs. 27.0; Qwen3-8B: 59.4 vs. 52.7; Qwen3.5-0.8B: 57.1 vs. 36.0; Qwen3.5-4B: 58.7 vs. 51.3; at 9B: 59.0 for Prism and 59.4 for Ours-9B vs. 54.7 for Clef-Flash). The lead is significant on 46 of the 60 backbone–collection pairs and never significantly reversed (*). The zero-shot subset tells the same story (e.g. 54.9 vs. 30.9 at 0.6B, 56.1 vs. 49.0 at 9B), so the gap is not owed to in-domain MS MARCO training; nor to proprietary data, since Ours-9B, trained on MS MARCO alone, matches Prism-9B. Decision fine-tuning barely moves the zero-shot base model up to 9B – its better mode lies between 3.5 points below and 3.2 above the base average, with significant differences in both directions on at most five collections (†) – whereas reranker fine-tuning adds 5 to 32 points. Only Kev-27B improves on its base (by 3.7, significant on eight collections); on that backbone, without a reranker fine-tune, it is the best system of the study (61.1), but only edges reranker fine-tunes a third its size (59.4).

Table 2. Other systems, same candidates (nDCG@10 ×100, means). †: includes published numbers [1,33]; –: not all collections available.

System	Type	DL	BEIR	Avg	ZS
BM25		39.3	47.1	43.2	45.2
RankGPT-4	listwise	–	56.9†	–	57.1†
RankZephyr-7B	listwise	61.4	53.3†	57.3†	51.7†
monoT5-3B	cross-encoder	60.4	54.6†	57.5†	52.5†
bge-reranker-v2-m3	cross-encoder	60.4†	52.8†	56.6†	52.2†
Set-Encoder-large	cross-encoder	63.6†	54.4	59.0†	53.8
Jev	System One, joint	62.7	58.2	60.5	57.2
Jev	System One, isolated	61.4	57.6	59.5	56.7
Laya	System One, isolated	47.3	40.7	44.0	43.7

Joint vs. isolated. Joint scoring helps only models trained on long states, and only slightly: Kev-27B gains 1.3 points and Jev 1.0 (Table 2; Kev-27B significantly on one collection), Kev-9B and Clef stay within 0.3, Kev-4B and Clef-Flash lose about 1.5 to 2, and Kev-0.6B, Kev-0.8B and Kev-8B lose 10.7 to 13.9, significantly on 6 to 10 collections (‡). Kev-0.8B and Kev-4B were trained on states of at most 7,552 tokens, Kev-27B on up to 32,768 [22] (Kev-8B's card states none), whereas a joint state has about 13k tokens on TREC DL and 40k or more on BEIR. Beyond that length a question is no longer tied to its passage: Kev-0.6B joint separates relevant from non-relevant passages at chance level on DL 2019/2020, its probabilities rising with the position in the state.

Context. The open Kev-27B (61.1) matches the closed Jev (60.5, Table 2); both lead established rerankers (Set-Encoder-large 59.0, monoT5-3B 57.5), as do Qwen3-Reranker-8B and Ours-9B (59.4). Laya (44.0) is barely above BM25 (43.2).

5 Are Decision Models Better Calibrated?

Calibrated probabilities are the selling point of decision models, and the zero-shot base models are indeed poorly calibrated (Table 3; ECE up to 0.750). Decision fine-tuning lowers their ECE on every backbone – but so does reranker fine-tuning, and against the reranker fine-tune of the same backbone the decision fine-tune has no consistent edge: it is worse at Qwen3-0.6B (0.579 vs. 0.511) and Qwen3.5-0.8B (0.357 vs. 0.243), on par at Qwen3.5-4B (0.216 vs. 0.218) and Qwen3-8B (0.243 vs. 0.257), and better only at Qwen3.5-9B against Prism (Clef-Flash 0.140 vs. 0.242). On that backbone, Ours-9B is the best calibrated system of the study (0.111), below every decision model including the 27B ones (0.147–0.156) and Jev (0.224): calibration follows from a pointwise probabilistic loss, not from decision fine-tuning. A count test – the summed probabilities of a query's judged candidates as an estimate of how many are relevant – agrees: decision and reranker fine-tune tie at Qwen3.5-4B (11.5 each) and Qwen3-8B (12.8 vs. 12.9), the reranker fine-tune counts better at the two smallest backbones, and Ours-9B counts best of all systems (7.3; Clef 8.4, Kev-27B 8.5, Jev 11.3).

Table 3. Calibration and one cutoff, means over TREC DL 2019–2023. Per backbone: base model, decision fine-tunes in their better-calibrated mode (j/i: joint/isolated), reranker fine-tunes. N: native; T, P: temperature/Platt scaling fitted on MS MARCO dev. F_1@.5 (×100): set kept at $p \geq 0.5$; Regret: F_1 at each collection's best cutoff minus F_1@.5; $|S|$: passages kept at 0.5 (of 100). Count: mean absolute error of the summed probabilities against the number of relevant passages (always guessing the median count: 8.6).

	ECE			F_1@.5				
System	N	T	P	N	P	Regret	$\|S\|$	Count
Qwen3-0.6B								
Qwen3-0.6B	0.750	–	–	19.9	–	0.5	100.0	36.5
Kev-0.6B (i)	0.579	–	–	20.7	–	2.4	87.9	28.2
Qwen3-Reranker-0.6B	0.511	0.326	0.226	31.8	6.3	14.9	58.4	24.8
Qwen3-8B								
Qwen3-8B	0.383	–	–	37.8	–	3.8	43.5	17.5
Kev-8B (j)	0.243	–	–	36.0	–	4.6	33.7	12.8
Qwen3-Reranker-8B	0.257	0.237	0.227	46.3	7.4	9.0	32.2	12.9
Qwen3.5-0.8B								
Qwen3.5-0.8B	0.716	–	–	20.0	–	12.3	98.6	34.8
Kev-0.8B (i)	0.357	–	–	24.7	–	3.5	56.2	18.0
Prism-0.8B	0.243	0.229	0.237	46.9	5.7	2.8	26.5	12.7
Qwen3.5-4B								
Qwen3.5-4B	0.296	–	–	40.3	–	7.1	38.0	14.7
Kev-4B (i)	0.216	–	–	41.0	–	4.6	30.5	11.5
Prism-4B	0.218	0.202	0.238	51.0	5.7	2.8	23.1	11.5

	ECE			F_1@.5				
System	N	T	P	N	P	Regret	$\|S\|$	Count
Qwen3.5-9B								
Qwen3.5-9B	0.283	–	–	42.1	–	7.4	36.8	14.1
Kev-9B (j)	0.237	–	–	41.0	–	4.6	32.2	12.2
Clef-Flash (j)	0.140	–	–	46.6	–	2.6	21.4	8.9
Prism-9B	0.242	0.233	0.238	50.7	5.7	4.0	26.0	12.3
Ours-9B	0.111	–	–	46.6	–	6.4	8.7	7.3
Qwen3.8-27B								
Qwen3.8-27B	0.166	–	–	52.2	–	2.4	24.2	8.9
Kev-27B (j)	0.156	–	–	55.1	–	2.8	22.8	8.5
Clef (j)	0.147	–	–	52.5	–	3.1	23.3	8.4
Other systems								
Jev (j)	0.224	–	–	49.9	–	6.2	29.1	11.3
monoT5-3B	0.312	0.257	0.232	42.0	5.8	9.9	35.6	15.2
bge-reranker-v2-m3	0.231	0.226	0.236	48.2	5.7	3.8	29.3	12.1
Set-Encoder-large	0.495	0.250	0.236	31.6	5.7	21.1	52.2	24.3

One cutoff for every collection. Keeping every candidate with $p \geq 0.5$ (e.g. to decide which passages reach a generator), the best reranker fine-tune yields the better set on every backbone where both exist (F_1@.5, e.g. Prism-4B 51.0 vs. Kev-4B 41.0, Prism-9B 50.7 vs. Clef-Flash 46.6). A low regret (0.5 close to each collection's best cutoff) is no privilege of decision models: Prism (2.8–4.0), bge (3.8) and the untuned Qwen3.8-27B (2.4) match Clef-Flash (2.6), Kev-27B (2.8) and Clef (3.1), and beat Jev (6.2); only rerankers whose scores were never meant as probabilities fall far behind (Set-Encoder 21.1). Low ECE alone is not enough: Ours-9B is underconfident on TREC DL, keeps only 8.7 passages per query and has a regret of 6.4.

Calibrated after the fact. Fitted on MS MARCO dev, temperature scaling lowers every reranker's ECE (e.g. Qwen3-Reranker-0.6B 0.511 to 0.326) but none gets below 0.20; Platt scaling brings all to about 0.23 but collapses the set at 0.5 (F_1@.5 5.7–7.4), since MS MARCO dev judges about one passage per query. Post-hoc scaling is no substitute for a calibrated model – but that model need not be a decision model, as Ours-9B shows.

6 Conclusion

On the same backbone, reranker fine-tuning beats decision fine-tuning wherever both exist (RQ1), and decision models are not better calibrated than reranker fine-tunes (RQ2): the probabilities that set them apart come from the training loss and are available to any reranker fine-tune. *Limitations*: no 27B reranker fine-tune exists yet; the fine-tunes saw different training data, which the zero-shot subset only partly controls; literature numbers are not recomputed or tested; calibration is measured on TREC DL only.

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